

Lecture 9. Extremal Set Theory

- extremal problems for set systems / set families / hypergraphs

$$\mathcal{F} \subseteq 2^{[n]}$$

k -uniform set system / hypergraph : $\forall F \in \mathcal{F}, |F| = k$

- theme sets interact $\left\{ \begin{array}{l} \bullet \text{ disjointness / intersect? } A \cap B = \emptyset? \\ \bullet \text{ containment } A \subseteq B? \end{array} \right.$

1. Sperner's thm

Def A partial order set (poset)

$(P, <)$
↑ set ↑ relation

- reflexive $\forall x \in P, x < x$

- antisymmetric $\begin{array}{l} x < y \\ y < x \end{array} \Rightarrow x = y$

- transitive $\begin{array}{l} x < y \\ y < z \end{array} \Rightarrow x < z$

If $x < y$ or $y < x$, we say x, y are comparable in $(P, <)$.
(in $(P, <)$)

- $A \subseteq P$ is antichain if elements in A are pairwise incomparable (i.e. $\forall a, b \in A, \begin{array}{l} a \not< b \\ b \not< a \end{array}$)

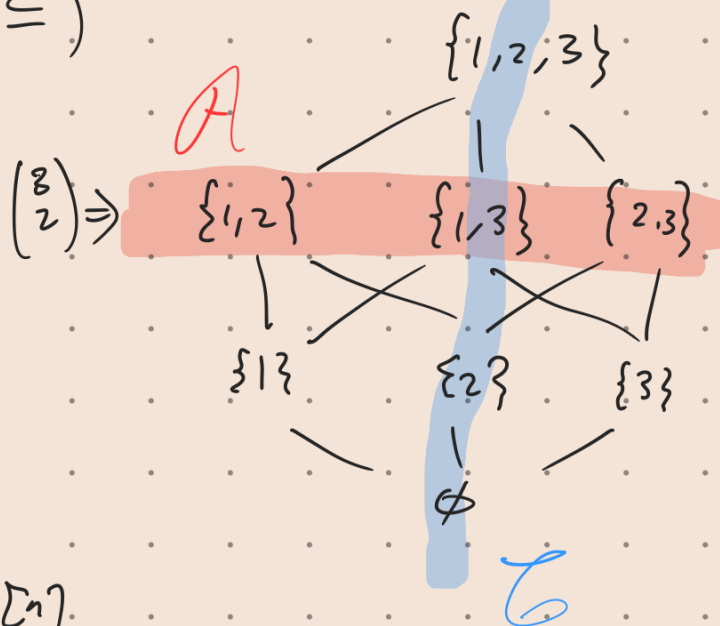
- $C \subseteq P$ chain if $\text{---} " \text{---} C \text{---}$
 $\text{---} " \text{---}$ comparable

Ex Boolean poset $(2^{[n]}, \subseteq)$

• antichain $A \subseteq 2^{[n]}$

$\forall S, T \in A, S \not\subseteq T$
(forbid containment)

• chain $S_1 \subseteq S_2 \subseteq \dots \subseteq S_t$



Thm 1.1 (Sperner 1928) $\mathcal{F} \subseteq 2^{[n]}$

if $\mathcal{F}$ is an antichain $\Rightarrow |\mathcal{F}| \leq \binom{n}{\lfloor n/2 \rfloor}$

$e(\mathcal{F})$
 middle level of Boolean poset

(i.e. the middle level is the largest antichain)

Bollobás set-pairs
ineq

HW

LYM ineq

HW

Sperner's thm

Thm 1.2 (LYM ineq) $\forall$ antichain $\mathcal{F} \subseteq 2^{[n]}$,

$$\sum_{F \in \mathcal{F}} \frac{1}{\binom{n}{|F|}} \leq 1$$

2: Bollobás set-pairs ineq

Thm 2.1 Given sets $A_1, \dots, A_m, B_1, \dots, B_m$, if

• $A_i \cap B_i = \emptyset, \forall i \in [m]$

• $A_i \cap B_j \neq \emptyset, \forall i \neq j$

$$\Rightarrow \sum_{i \in [m]} \frac{1}{\binom{|A_i| + |B_i|}{|A_i|}} \leq 1$$

In particular, $\forall i \in [m], |A_i| = a, |B_i| = b$

$$\Rightarrow m \leq \binom{a+b}{a}$$

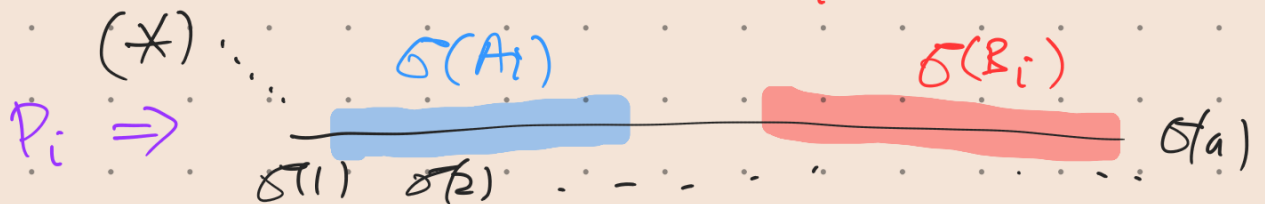
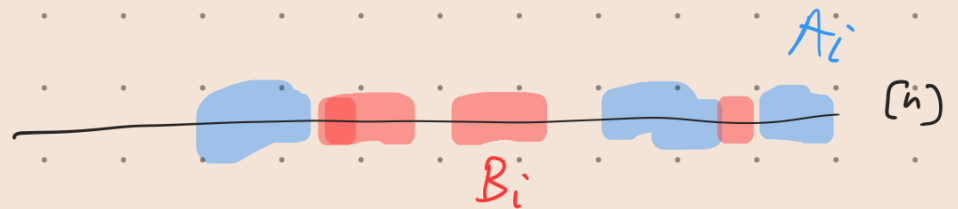
Pf: (double-counting)

• Assume $[n] = \bigcup_{i \in [m]} (A_i \cup B_i)$.

• Take a ^{uniform} random permutation $\sigma \in S_n$.

• For each $i \in [m]$, define event

$P_i = A_i$ comes in σ before B_i



Event P_i depends only on relative position of $\sigma(A_i), \sigma(B_i)$ in $\sigma(A_i \cup B_i)$

$$P(P_i) = \frac{|A_i|! \cdot |B_i|!}{(|A_i| + |B_i|)!} = \frac{1}{\binom{|A_i| + |B_i|}{|A_i|}}$$

Claim All P_i events, $i \in [m]$ are disjoint.

(Claim $\Rightarrow$ then)

$$1 \geq \Pr\left(\bigcup_{i \in [m]} P_i\right) \stackrel{\text{Claim}}{\geq} \sum_{i \in [m]} \Pr(P_i) = \sum_i \frac{1}{\binom{|A_i| + |B_i|}{|A_i|}}$$

Pf of Claim $i \neq j \in [m]$. Suppose P_i occurs.

(NTS P_j does NOT happen)

$$\max \sigma(A_i) < \min \sigma(B_i) \dots \dots \dots (*)$$

$$\min \sigma(B_j) \leq A_i \cap B_j \neq \emptyset$$

$$\leq \max \sigma(A_j) \quad A_j \cap B_i \neq \emptyset$$

$\Rightarrow P_j$ not happening $\square$

Ex 2.2 Give an example showing that Thm 2.1 is tight.

2.1 Skewed version (Lovász, Alon-Kalai)

Thm 2.3 Given $A_1, \dots, A_m, B_1, \dots, B_m$, if

- $|A_i| \leq a, |B_i| \leq b \quad \forall i \in [m]$
- $A_i \cap B_i = \emptyset, \quad \forall i \in [m]$
- $A_i \cap B_j \neq \emptyset, \quad \forall j > i$

$$\Rightarrow m \leq \binom{a+b}{a}$$

Remark: For the skewed version, no combinatorial pf is known.

• field K , vect sp. V over K

• exterior algebra $\bigwedge V = \bigotimes V / x \otimes x$

$$x \wedge x = 0 \quad \forall x$$

- k^{th} exterior power of V

$\wedge^k V =$ vect. sp. of $\wedge V$ spanned by

$$x_1 \wedge x_2 \wedge \dots \wedge x_k, \quad x_i \in V, \quad i \in [k]$$

- $x_1 \wedge x_2 \wedge \dots \wedge x_k = 0$ if $x_i = x_j$ for some $1 \leq i < j \leq k$.

• $\dim V = n$

• $\dim (\wedge^k V) = \binom{n}{k}$

pf (Thm 2.3) May assume $|A_i| = a$.

$[n] = \bigcup_{i \in [m]} (A_i \cup B_i)$ ground set

- Take $V = \mathbb{R}^{a+b}$, take $v_1, \dots, v_n \in V$

in general position ($\forall$ subset of $\leq a+b$ of v_i 's)
they are linearly indep.

- For each $i \in [m]$, define

$$y_i = \bigwedge_{j \in A_i} v_j, \quad z_i = \bigwedge_{k \in B_i} v_k$$

$\in \wedge^a V$

all distinct
↓ ↓ ↓
 $x_1 \wedge x_2 \wedge x_3 \wedge x_4$
 $\neq 0$

- v_i 's are in general position

$$A_i \cap B_i = \emptyset \Rightarrow y_i \wedge z_i \neq 0 \dots (1)$$

$$\forall j > i, A_i \cap B_j \neq \emptyset \Rightarrow y_i \wedge z_j = 0 \dots (2)$$

Suffices to show $y_1, \dots, y_m$ are linearly indep.

$$\Rightarrow m \leq \dim(\wedge^a \mathbb{R}^{atb}) = \binom{atb}{a}$$

• Suppose $\forall \alpha_1 y_1 + \dots + \alpha_m y_m = 0$

WTS all $\alpha_i = 0$

$$0 = 0 \wedge Z_m = (\alpha_1 y_1 + \dots + \alpha_m y_m) \wedge Z_m$$

$$\stackrel{(2)}{=} \alpha_m (y_m \wedge Z_m)$$

By (1) $\Rightarrow \alpha_m = 0$. Repeat $\dots \alpha_i = 0 \forall i$. 

• Sperner's thm

• Bollobás's set-pair ineq / Skewed version.

Applications

1) Littlewood - Offord problem:

- bound the atom probability of Rademacher sum.
- used in bounding singularity prob. of random matrices.

Recall a Rademacher random variable $X = \begin{cases} 1 & \text{w/ prob } 1/2 \\ -1 & \dots 1/2 \end{cases}$

Thm (Erdős 45) Let $a_1, \dots, a_n \in \mathbb{Z} \setminus \{0\}$, and let $x_1, \dots, x_n$ be i.i.d. ^{independently identically distributed} Rademacher random variables.

$$\text{Then } \sup_{z \in \mathbb{Z}} \Pr\left(\sum_{i \in [n]} a_i x_i = z\right) \leq \frac{\binom{n}{\lfloor n/2 \rfloor}}{2^n}$$

Pf: • The total # choices = $|\{-1, 1\}^n| = 2^n$
 $(x_1, \dots, x_n)$

• Fix $z \in \mathbb{Z}$.

NTS $\# \left(\begin{array}{c} \text{choices of } (x_1, \dots, x_n) \text{ s.t.} \\ \sum a_i x_i = z \\ S \subseteq \{-1, 1\}^n \end{array} \right) \leq \binom{n}{\lfloor n/2 \rfloor}$

Idea: identify S as subset of $2^{[n]}$
 show that S is an antichain.

$(-1, -1, 1, 1, -1, 1) \in S$
 $(0, 0, 1, 1, 0, 1)$
 $\Downarrow$
 $\{3, 4, 6\}$ $[n] = 6$

$S \subseteq 2^{[n]}$

It suffices to show $(y_1, \dots, y_n) \in S \iff Y, i \in Y \iff y_i = 1$.

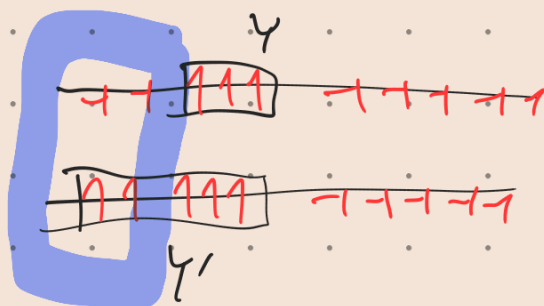
Claim S is an antichain. (Sperner $\Rightarrow$ desired bound)

Pf • May assume by symm. $a_i > 0, \forall i \in [n]$.

$(y_1, \dots, y_n) \in Y \subseteq Y' \in S$

$z = \sum a_i y_i = \sum a_i y_i'$

$0 = \sum (a_i y_i' - a_i y_i) = 2 \sum_{i \in Y' \setminus Y} a_i > 0$



$\geq \dots$

Rmk The bound is tight, considering

say all $a_i = 1$, $z = 0$.

2) LYM inequality.

Ex Bollobás's set pairs inequality $\Rightarrow$ LYM inequality.

3) Saturation number.

maximally H -free

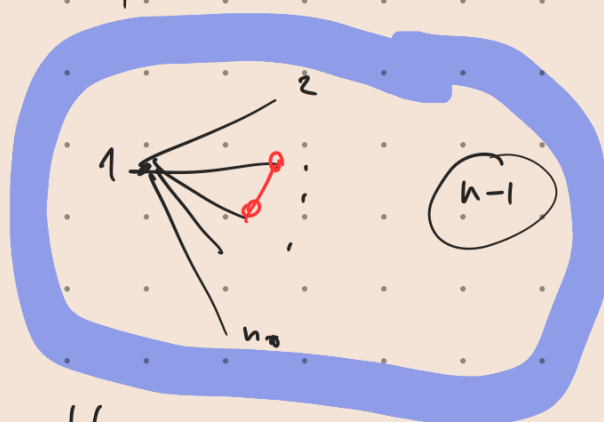
Given k -graph (k -uniform hypergraph) H ,

say k -graph G is H -saturated if

- H -free

- $G \cup e \geq H \quad \forall e \notin G$

Ex $k=2$ $H = \Delta$



Def: The saturation number of H

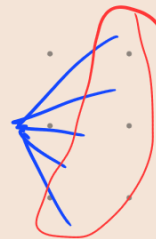
$\text{sat}(n, H) = \min \{ e(G) : G \text{ is } n\text{-vx } H\text{-saturated} \}$

$$\text{sat}(n, \Delta) \leq n-1$$

• Write $K_t^{(k)} = t$ -vx complete k -unif hypergraph

Ex 3.3 (Bollobás 65) $\forall n \geq t \geq k \geq 2$,

$$\text{sat}(n, K_t^{(k)}) = \binom{n}{k} - \binom{n-t+k}{k}$$



ex above for Δ , $k=2$, $t=3$

$$\binom{n}{2} - \binom{n-1}{2}$$